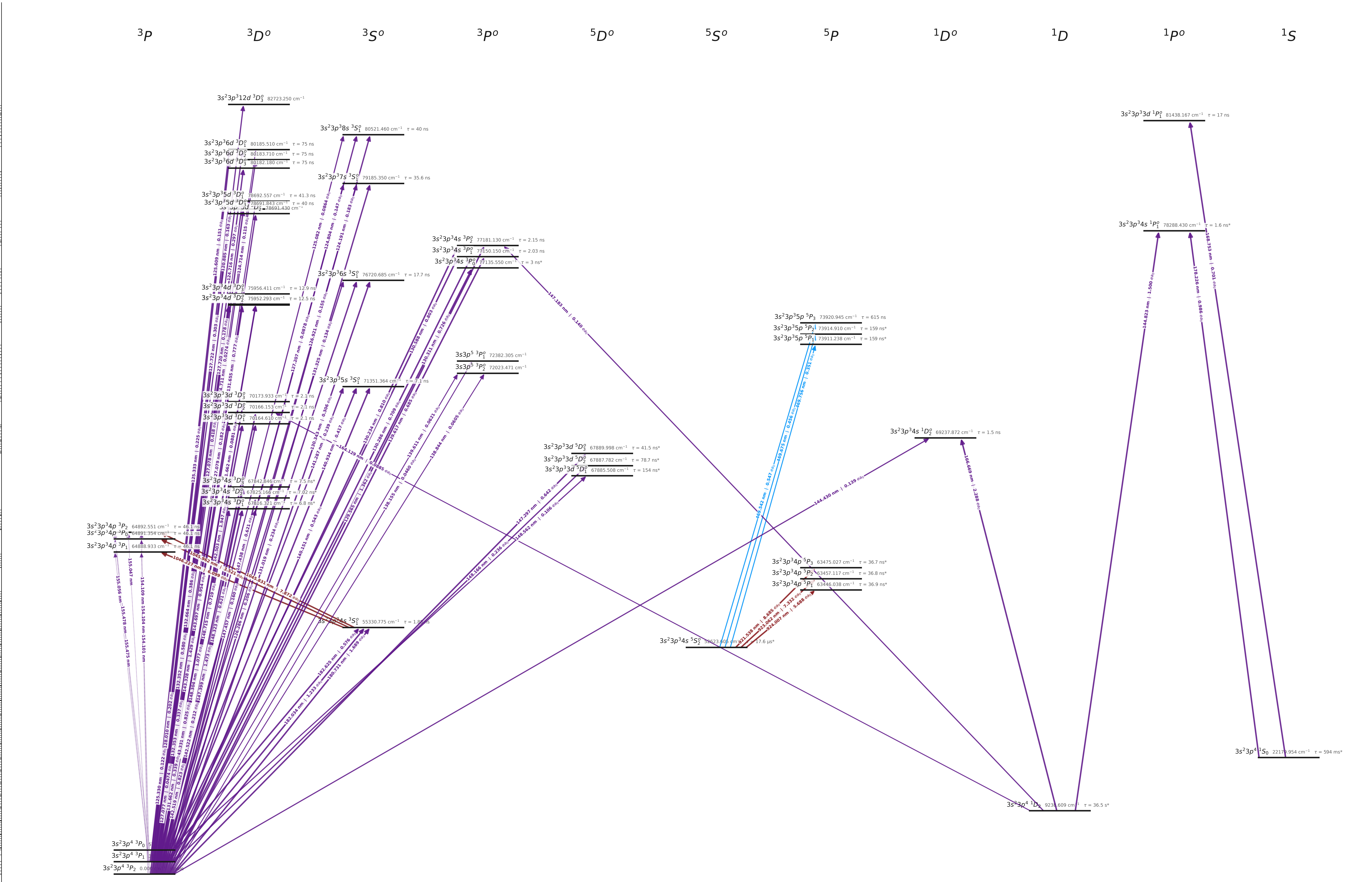


# Sulfur

S (S I). Strongest classified transitions below 2 μm (NIST): 47 levels, 81 lines.

S<sup>+</sup> ionisation limit 83559.17 cm<sup>-1</sup> = 10.36002 eV (119.676 nm)



## Key transitions

| Transition                                                                                                           | λ vac (nm) | λ air (nm) | ν (THz)         | ⟨J  er  ⟩  (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                         |
|----------------------------------------------------------------------------------------------------------------------|------------|------------|-----------------|-------------------------------|----------------------|----------------|------------|-------------------------------------------------------------|
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>2</sub> – 3s <sup>2</sup> 3p <sup>3</sup> 4s ³P <sub>2</sub> <sup>o</sup>    | 129.5653   | –          | 2313.8321       | 1.362                         | 3.457e+08            | 74.16          | 74.2       | 129.6 nm ground-state lines of the 1299 A multiplet used... |
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>2</sub> – 3s <sup>2</sup> 3p <sup>3</sup> 4s ³P <sub>1</sub> <sup>o</sup>    | 129.6174   | –          | 2312.9033       | 0.685                         | 1.456e+08            | 78.25          | 29.6       | 129.6 nm ground-state lines of the 1299 A multiplet used... |
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>2</sub> – 3s <sup>2</sup> 3p <sup>3</sup> 4p ³P <sub>2</sub>                 | 154.1009   | –          | 1945.429689(27) | –                             | –                    | 3.452          | –          | Doppler-free two-photon excitation                          |
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>2</sub> – 3s <sup>2</sup> 3p <sup>3</sup> 4p ³P <sub>0</sub>                 | 154.1037   | –          | 1945.393840(27) | –                             | –                    | 3.452          | –          | Doppler-free two-photon excitation                          |
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>2</sub> – 3s <sup>2</sup> 3p <sup>3</sup> 4p ³P <sub>1</sub>                 | 154.1095   | –          | 1945.321233(27) | –                             | –                    | 3.452          | –          | Doppler-free two-photon excitation                          |
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>1</sub> – 3s <sup>2</sup> 3p <sup>3</sup> 4p ³P <sub>2</sub>                 | 155.0472   | –          | 1933.55618(7)   | –                             | –                    | 3.452          | –          | Doppler-free two-photon excitation                          |
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>1</sub> – 3s <sup>2</sup> 3p <sup>3</sup> 4p ³P <sub>1</sub>                 | 155.0559   | –          | 1933.447755(27) | –                             | –                    | 3.452          | –          | Doppler-free two-photon excitation                          |
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>0</sub> – 3s <sup>2</sup> 3p <sup>3</sup> 4p ³P <sub>2</sub>                 | 155.4752   | –          | 1928.233794(27) | –                             | –                    | 3.452          | –          | Doppler-free two-photon excitation                          |
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>0</sub> – 3s <sup>2</sup> 3p <sup>3</sup> 4p ³P <sub>0</sub>                 | 155.4781   | –          | 1928.197819(27) | –                             | –                    | 3.452          | –          | Doppler-free two-photon excitation                          |
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>2</sub> – 3s <sup>2</sup> 3p <sup>3</sup> 4s ³S <sub>1</sub> <sup>o</sup>    | 180.7312   | –          | 1658.7749       | 1.689                         | 3.264e+08            | 84.88          | 61.2       | 180.731 nm                                                  |
| 3s <sup>2</sup> 3p <sup>3</sup> 4s ³S <sub>1</sub> <sup>o</sup> – 3s <sup>2</sup> 3p <sup>3</sup> 4p ³P <sub>2</sub> | 1045.8308  | 1045.5443  | 286.6548        | 7.872                         | 2.195e+07            | 3.452          | 100        | 1045.5-1045.9 nm near-infrared triplet                      |
| 3s <sup>2</sup> 3p <sup>3</sup> 4s ³S <sub>1</sub> <sup>o</sup> – 3s <sup>2</sup> 3p <sup>3</sup> 4p ³P <sub>0</sub> | 1045.9618  | 1045.6753  | 286.6189        | 3.521                         | 2.195e+07            | 3.452          | 100        | 1045.5-1045.9 nm near-infrared triplet                      |
| 3s <sup>2</sup> 3p <sup>3</sup> 4s ³S <sub>1</sub> <sup>o</sup> – 3s <sup>2</sup> 3p <sup>3</sup> 4p ³P <sub>1</sub> | 1046.2267  | 1045.9401  | 286.5464        | 6.099                         | 2.194e+07            | 3.452          | 100        | 1045.5-1045.9 nm near-infrared triplet                      |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

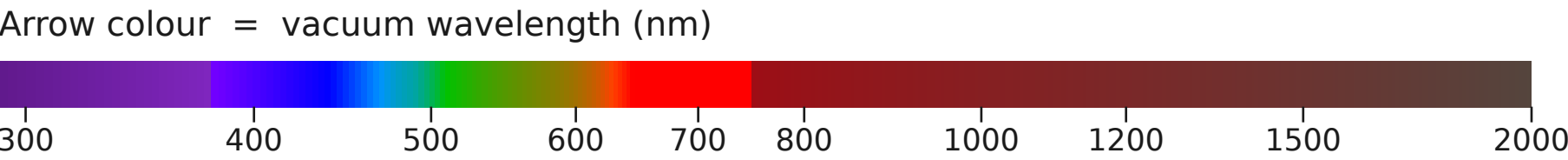
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||⟩| in ea<sub>0</sub>

Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.

Level caption: NIST energy and measured lifetime (where one exists).



## ³³S hyperfine structure (I = 3/2)

| Level                                           | A (MHz) | B (MHz) | Shift of each F level from the centre of gravity (MHz)     |
|-------------------------------------------------|---------|---------|------------------------------------------------------------|
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>2</sub> | 130.090 | -31.600 | F=1/2: -613.05 F=3/2: -390.27 F=5/2: -45.30 F=7/2: +382.37 |
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>1</sub> | -4.620  | 15.800  | F=1/2: +31.30 F=3/2: -11.18 F=5/2: -2.98                   |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition                                                                                           | λ vac (nm) | Isotopes | Shift (MHz) |
|------------------------------------------------------------------------------------------------------|------------|----------|-------------|
| 3s <sup>2</sup> 3p <sup>4</sup> ³P <sub>2</sub> – 3s <sup>2</sup> 3p <sup>3</sup> 4p ³P <sub>2</sub> | 154.101    | 34-32    | 537(12)     |

Shift = ν(first isotope) – ν(second isotope).